

Dynamics: Planar Kinematics of Rigid Body

- ☐ ABSOLUTE MOTION ANALYSIS
- ☐ RELATIVE MOTION ANALYSIS
- ☐ PROBLEMS



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ME101 (Spring 2017)

Planar Kinematics of Rigid Body

- To investigate rigid-body translation and angular motion about a fixed axis.
- To study planar motion using an absolute motion analysis.
- To provide a relative motion analysis of velocity and acceleration using a translating frame of reference.
- To provide a relative-motion analysis of velocity and acceleration using a rotating frame of reference.

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RIGID BODY TRANSLATION

POSITION, VELOCITY, & ACCELERATION

Relating the variables, s, v, a , & t

$$v = \frac{ds}{dt} \quad a = \frac{dv}{dt} \quad a ds = v dv$$

$$a_c dt = dv \quad v = \frac{ds}{dt} = v_0 + a_c t \quad v dv = a_c ds$$

$$\int_{v_0}^v dv = \int_0^t a_c dt \quad \int_{s_0}^s ds = \int_0^t (v_0 + a_c t) dt \quad \int_{v_0}^v v dv = \int_{s_0}^s a_c ds$$

$$v = v_0 + a_c t \quad s = s_0 + v_0 t + \frac{1}{2} a_c t^2 \quad v^2 = v_0^2 + 2a_c(s - s_0)$$

Constant Acceleration Constant Acceleration Constant Acceleration

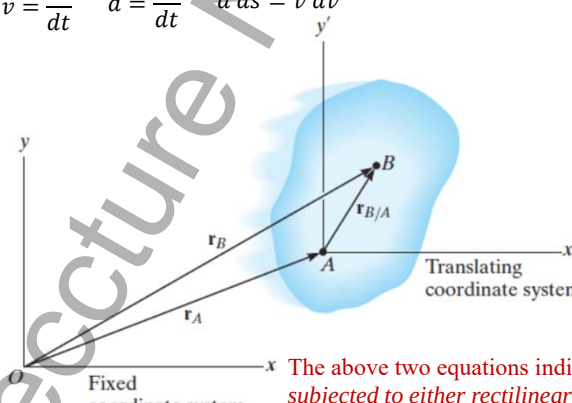
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RIGID BODY TRANSLATION

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$$v = \frac{ds}{dt} \quad a = \frac{dv}{dt} \quad a ds = v dv$$


The position of B with respect to A is denoted by the *relative-position vector* $\mathbf{r}_{B/A}$ ("r of B with respect to A").

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$$

Taking the first derivative,

$$\frac{d\mathbf{r}_B}{dt} = \frac{d\mathbf{r}_A}{dt} + \frac{d\mathbf{r}_{B/A}}{dt}$$

$$\mathbf{v}_B = \mathbf{v}_A + \frac{d\mathbf{r}_{B/A}}{dt}$$

$$\mathbf{v}_B = \mathbf{v}_A$$

Taking the derivative again,

$$\mathbf{a}_B = \mathbf{a}_A$$

The above two equations indicate that *all points in a rigid body subjected to either rectilinear or curvilinear translation move with the same velocity and acceleration.*

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ROTATION ABOUT A FIXED AXIS

ANGULAR DISPLACEMENT, VELOCITY, & ACCELERATION

Relating the variables, θ , ω , α , & t

$\omega = \frac{d\theta}{dt}$

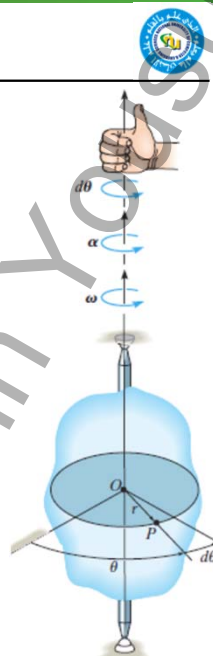
$\alpha = \frac{d\omega}{dt}$

$\alpha = \frac{d^2\theta}{dt^2}$

$\alpha d\theta = \omega d\omega$

$$\begin{aligned}\omega &= \omega_0 + \alpha_c t \\ \theta &= \theta_0 + \omega_0 t + \frac{1}{2} \alpha_c t^2 \\ \omega^2 &= \omega_0^2 + 2\alpha_c(\theta - \theta_0)\end{aligned}$$

Constant Angular Acceleration



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$\omega = \frac{d\theta}{dt}$

$\alpha = \frac{d\omega}{dt}$

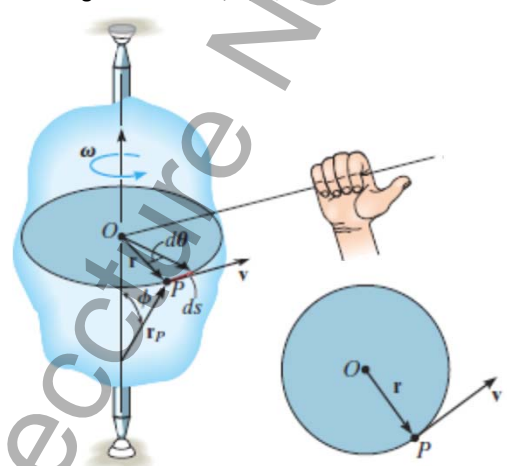
$\alpha d\theta = \omega d\omega$

$\alpha = \frac{d^2\theta}{dt^2}$

$ds = r d\theta$

$\frac{ds}{dt} = v = r \frac{d\theta}{dt} = r\omega$

$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}_P$



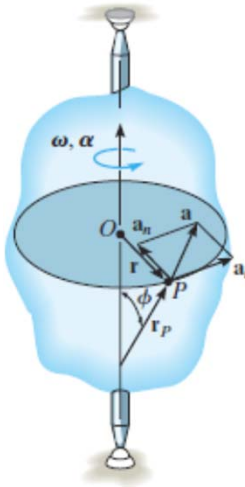
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ROTATION ABOUT A FIXED AXIS

ANGULAR DISPLACEMENT, VELOCITY, & ACCELERATION

Relating the variables, θ , ω , α , & t



Taking first derivative of

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}_P$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(\boldsymbol{\omega} \times \mathbf{r}_P)$$

$$\mathbf{a} = \frac{d\boldsymbol{\omega}}{dt} \times \mathbf{r}_P + \boldsymbol{\omega} \times \frac{d\mathbf{r}_P}{dt}$$

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r}_P + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_P)$$

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt}$$

$$\alpha d\theta = \omega d\omega$$

$$\alpha = \frac{d^2\theta}{dt^2}$$

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}_P$$

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r} - \omega^2 \mathbf{r}$$

$$\mathbf{a} = \mathbf{a}_t + \mathbf{a}_n$$

$$a_t = \alpha r$$

$$a_n = \omega^2 r$$

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Procedure for Analysis

ANGULAR MOTION

- Establish the positive sense of rotation about the axis of rotation and show it alongside each kinematic equation as it is applied.
- If a relation is known between any two of the four variables α , ω , θ and t , then a third variable can be obtained by using one of the following kinematic equations which relates all three variables

$$\omega = \frac{d\theta}{dt}, \quad \alpha = \frac{d\omega}{dt}, \quad \alpha d\theta = \omega d\omega$$
- If the body's angular acceleration is constant, then the following equations can be used:

$$\omega = \omega_0 + \alpha_c t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha_c t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$$

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Procedure for Analysis

MOTION OF POINT P

- In most cases the velocity of P and its two components of acceleration can be determined from the scalar equations

$$v = \omega r, \quad a_t = \alpha r, \quad a_n = \omega^2 r$$

- If the geometry of the problem is difficult to visualize, the following vector equations should be used:

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}_P = \boldsymbol{\omega} \times \mathbf{r}$$

$$\mathbf{a}_t = \boldsymbol{\alpha} \times \mathbf{r}_P$$

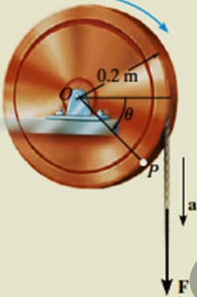
$$\mathbf{a}_n = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_P) = -\omega^2 \mathbf{r}$$

- Here $\mathbf{r}_P$ is directed from any point on the axis of rotation to point P , whereas $\mathbf{r}$ lies in the plane of motion of P .

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A cord is wrapped around a wheel in Fig. 16-5, which is initially at rest when $\theta = 0$. If a force is applied to the cord and gives it an acceleration $a = (4t) \text{ m/s}^2$, where t is in seconds, determine, as a function of time, (a) the angular velocity of the wheel, and (b) the angular position of line OP in radians.

$$(a_P)_t = \alpha r$$

$$(4t) \text{ m/s}^2 = \alpha (0.2 \text{ m})$$

$$\alpha = (20t) \text{ rad/s}^2$$

$$\alpha = \frac{d\omega}{dt} = (20t) \text{ rad/s}^2$$

$$\int_0^\omega d\omega = \int_0^t 20t \, dt$$

$$\omega = 10t^2 \text{ rad/s}$$

$$\frac{d\theta}{dt} = \omega = (10t^2) \text{ rad/s}$$

$$\int_0^\theta d\theta = \int_0^t 10t^2 \, dt$$

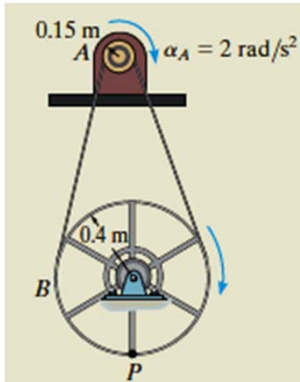
$$\theta = 3.33t^3 \text{ rad}$$

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The motor shown in the photo is used to turn a wheel and attached blower contained within the housing. The details of the design are shown in Fig. 16-6a. If the pulley A connected to the motor begins to rotate from rest with a constant angular acceleration of $\alpha_A = 2 \text{ rad/s}^2$, determine the magnitudes of the velocity and acceleration of point P on the wheel, after the pulley has turned two revolutions. Assume the transmission belt does not slip on the pulley and wheel.



$$\theta_A = 2 \text{ rev} \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) = 12.57 \text{ rad}$$

$$\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0) \quad \omega_A^2 = 0 + 2(2 \text{ rad/s}^2)(12.57 \text{ rad} - 0)$$

$$\omega_A = 7.090 \text{ rad/s}$$

$$v = \omega_A r_A = \omega_B r_B \quad 7.090 \text{ rad/s} (0.15 \text{ m}) = \omega_B (0.4 \text{ m})$$

$$\omega_B = 2.659 \text{ rad/s}$$

$$a_t = \alpha_A r_A = \alpha_B r_B \quad 2 \text{ rad/s}^2 (0.15 \text{ m}) = \alpha_B (0.4 \text{ m})$$

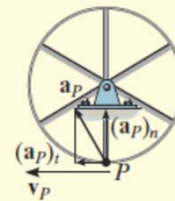
$$\alpha_B = 0.750 \text{ rad/s}^2$$

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Motion of P .

$$v_P = \omega_B r_B = 2.659 \text{ rad/s} (0.4 \text{ m}) = 1.06 \text{ m/s}$$

$$(a_P)_t = \alpha_B r_B = 0.750 \text{ rad/s}^2 (0.4 \text{ m}) = 0.3 \text{ m/s}^2$$

$$(a_P)_n = \omega_B^2 r_B = (2.659 \text{ rad/s})^2 (0.4 \text{ m}) = 2.827 \text{ m/s}^2$$

$$a_P = \sqrt{(0.3 \text{ m/s}^2)^2 + (2.827 \text{ m/s}^2)^2} = 2.84 \text{ m/s}^2$$

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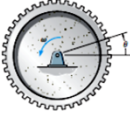
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322

CHAPTER 16 PLANAR KINEMATICS OF A RIGID BODY

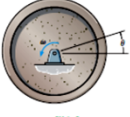
FUNDAMENTAL PROBLEMS

F16-1. When the gear rotates 20 revolutions, it achieves an angular velocity of $\omega = 30 \text{ rad/s}$, starting from rest. Determine its constant angular acceleration and the time required.



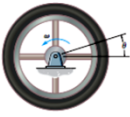
F16-1

F16-2. The flywheel rotates with an angular velocity of $\omega = (0.005\theta^2) \text{ rad/s}$, where θ is in radians. Determine the angular acceleration when it has rotated 20 revolutions.



F16-2

F16-3. The flywheel rotates with an angular velocity of $\omega = (4\theta^{3/2}) \text{ rad/s}$, where θ is in radians. Determine the time it takes to achieve an angular velocity of $\omega = 150 \text{ rad/s}$. When $t = 0$, $\theta = 0$.



F16-3

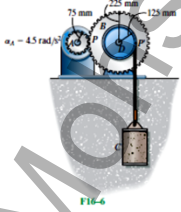
F16-4. The bucket is hoisted by the rope that wraps around a drum wheel. If the angular displacement of the wheel is $\theta = (0.5t^2 + 15t) \text{ rad}$, where t is in seconds, determine the velocity and acceleration of the bucket when $t = 3 \text{ s}$.



F16-4

F16-5. A wheel has an angular acceleration of $\alpha = (0.5\theta) \text{ rad/s}^2$, where θ is in radians. Determine the magnitude of the velocity and acceleration of a point P located on its rim after the wheel has rotated 2 revolutions. The wheel has a radius of 0.2 m and starts from rest.

F16-6. For a short period of time, the motor turns gear A with a constant angular acceleration of $\alpha_A = 4.5 \text{ rad/s}^2$, starting from rest. Determine the velocity of the cylinder and the distance it travels in three seconds. The cord is wrapped around pulley D which is rigidly attached to gear B .



F16-6

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